Comp 3004

Team Project: Traceability Matrix

ID	Requirement	Related Use Case	Implemented / Fulfilled By	Test	Description
1	GUI resembles AED Plus's design and inputs	N/A	Mainwindow.ui	Run simulation and observe GUI	The base AED Plus is replicated showing treatment progressions steps. A series of testing buttons have been set up as well using QT's UI framework to facilitate then user's interaction with the simulation.
2	Battery Display showing current charge status	N/A	Mainwindow.ui, AED, Battery	Observe battery indicator on top right of simulated unit	QT's LCDScreen is used to display current battery charge.
3	Electrode Placement Status	3: 1	Mainwindow.ui, AED	Click "place electrodes" button amd observe device feedback	AED keeps track of electrode placement and gives feedback on progressing to next states once placed.
4	Correct Adult/Child electrode is used	3: 3	Mainwindow.ui, AED, <Awaiting Implementation>	On active unit enter patient's age and observe how either adult or child pads are selectable	
5	Display of ECG	N/A	Mainwindow.ui, AED, ECG	Power on unit, place electrodes, and observe ECG waveform	Unit shows simulated wavelengths for different rhythms, while unit is powered on, and electrodes are placed.
6	Unit detects different Cardiac Atthytmia states	4: 3	Mainwindow.ui, AED, ECG, Display	On Active and connected unit click the rhythm testing buttons to observe	ECG keeps track of which rhythm the patient is in and reports back if it is shockable or not. The AED forwards that

				waveforms on the ECG and receive input on shockability	information to Display along with device information to display the relevant information.
7	Unit applies shock if required	4: 4a, 5: 2	Mainwindow.ui, AED, ECG	On active and connected unit in a test state with a shockable rhythm click the shock button	If ECG reports a shockable state, then the unit applies a shock and checks if the state has changed.
8	Unit tracks CPR status	4: 4b	Mainwindow.ui, AED, ECG, Display	On active and connected unit in a test state with a non-shockable rhythm apply CPR at varying depths	When the ECG detects a non-shockable rhythm the shock button is disabled and the unit advises CPR be applied to the patient. Further the unit tracks the depth at which CPR is performed and only advances if it is sufficiently deep.
9	Unit battery drains passively while active	N/A	Mainwindow.ui, AED, Battery	Power on unit and observe how charge drains over time	AED informs the Battery to decrease the charge based on a timer after the power on status has been applied.
10	Unit battery drains by a substantial amount on shock	N/A	Mainwindow.ui, AED, Battery, ECG	On active and connected unit with a shockable rhythm press shock button and observe battery level decrease	AED informs Battery to drain by large amount when the shock button is pressed while all the conditions to apply a shock are true.
11	Unit only powers on with sufficient charge	2: 1a	Mainwindow.ui, AED, Battery	On inactive unit with no charge attempt to power on unit	On power up AED asks battery if charge level is sufficient for power on, and only proceeds if it is.

12	Unit shock require a set amount of charge to proceed	N/A	Mainwindow.ui AED, Battery	On active and connected unit attempt to apply a shock with less than the required amount of Battery charge.	AED asks Battery for the charge level when attempting to perform an action and if that level is insufficient then informs the user to replace the Battery.
13	Unit battery can be replaced	N/A	Mainwindow.ui, AED, Battery	On inactive unit click replace button and observe how the battery level is reset to max on power on.	AED tells Battery to set charge to max so long as the unit is powered off.
14	Patient status is constant between device on and off cycles	N/A	Mainwindow.ui, AED, ECG	On active and connected unit observe how the testing state applied is remembered even if unit is powered off or electrodes disconnected	The ECG holds the current patient status until either a treatment changes that status, or a rhythm testing button is clicked.
15	The AED Plus device initiates a self-test to ensure that the device is functioning correctly and prepared for use.	2: 2	Mainwindow.ui, AED	Run simulation and observe GUI.	AED performs self-test and provides a boolean value (true/false) to determine if it is successful or unsuccessful. If unsuccessful user is prompted to change the batteries.
16	Unit has LED prompts showing what the current treatment step is	N/A	Mainwindow.ui, AED	Run simulation and follow prompts, observe how LED's illustrate state	As the program advances through the user interacting with QT UI elements or by timers updating the current state of the program corresponds with QLabel's representing LEDs.